Sample Paper 14

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

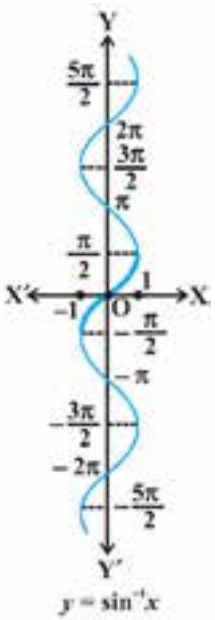
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the value of y when $x = -1$ in the graph of $y = \cos^{-1}x$?



(a) 0 (b) $\pi/2$

(c) π (d) –1

2. For what values of x, function $f(x) = x^4 - 4x^3 + 4x^2 + 40$ is monotonic decreasing?

(a) $0 < x < 1$ (b) $1 < x < 2$

(c) $2 < x < 3$ (d) $4 < x < 5$

3. The solution of $\frac{dy}{dx} = \frac{ax+g}{by+f}$ represents a circle, when

(a) $a = b$ (b) $a = -b$

(c) $a = -2b$ (d) $a = 2b$

4. If $\int_0^a f(2a-x)\,dx = m$ and $\int_0^a f(x)\,dx = n$, then $\int_0^{2a} f(x)\,dx$ is equal to

(a) $2m + n$ (b) $m + 2n$

(c) $m - n$ (d) $m + n$

5. Find the area of a curve $xy = 4$, bounded by the lines $x = 1$ and $x = 3$ and X-axis.

(a) $\log 12$ (b) $\log 64$

(c) $\log 81$ (d) $\log 27$

6. Order of the equation $\left(1 + 5\frac{dy}{dx}\right)^{3/2} = 10\frac{d^3y}{dx^3}$ is

(a) 2 (b) 3

(c) 1 (d) 0

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7. The figure formed by four points $\hat{i}+\hat{j}+\hat{k}$, $2\hat{i}+3\hat{j}$, $3\hat{i}-5\hat{j}-2\hat{k}$, $\hat{k}-\hat{j}$ is a

(a) Parallelogram

(b) Rectangle

(c) Trapezium

(d) Square

8. If $P(A \cup B)=0.83$, $P(A)=0.3$ and $P(B)=0.6$, then the events will be

(a) Dependent

(b) Independent

(c) Cannot say anything

(d) None of the above

9. Find the equation of the line which passes through the point $(1,2,3)$ and is parallel to the vector $3\hat{i}+2\hat{j}-2\hat{k}$.

(a) $\left(\hat{i}+\hat{j}+3\hat{k}\right)+\lambda\left(3\hat{i}+2\hat{j}-2\hat{k}\right)$

(b) $\left(\hat{i}+2\hat{j}+3\hat{k}\right)+\lambda\left(3\hat{i}+2\hat{j}-2\hat{k}\right)$

(c) $\left(\hat{i}+2\hat{j}+5\hat{k}\right)+\lambda\left(3\hat{i}+2\hat{j}-2\hat{k}\right)$

(d) $\left(\hat{i}+2\hat{j}+3\hat{k}\right)+\lambda\left(5\hat{i}+2\hat{j}-2\hat{k}\right)$

10. $\int \sqrt{1+\cos x}\,dx$ is equal to

(a) $2\sin\left(\frac{x}{2}\right)+C$ (b) $\sqrt{2}\sin\left(\frac{x}{2}\right)+C$

(c) $2\sqrt{2}\sin\left(\frac{x}{2}\right)+C$ (d) $\frac{1}{2}\sin\left(\frac{x}{2}\right)+C$

11. The family of curves $y=e^{a\sin x}$, where a is an arbitrary constant is represented by the differential equation

(a) $\log y=\tan x\frac{dy}{dx}$ (b) $y\log y=\tan x\frac{dy}{dx}$

(c) $y\log y=\sin\frac{dy}{dx}$ (d) $\log y=\cos x\frac{dy}{dx}$

12. If a line makes angles 90°, 60° and θ with X, Y and Z-axis respectively, where θ is acute angle, then find θ.

(a) 30° (b) 60°

(c) 45° (d) 90°

13. The area bounded by the curve $y=\frac{1}{2}x^2$, the X-axis and the ordinate $x=2$ is

(a) $\frac{1}{3}$ sq unit (b) $\frac{2}{3}$ sq unit

(c) 1 sq unit (d) $\frac{4}{3}$ sq unit

14. If $A=[a_{ij}]_{2\times 2}$, where $a_{ij}=i+j$, then A is equal to

(a) $\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix}$ (b) $\begin{bmatrix}1 & 2\\ 1 & 2\end{bmatrix}$

(c) $\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ (d) $\begin{bmatrix}2 & 3\\ 3 & 4\end{bmatrix}$

15. Discuss the continuity of the function $f(x)=\sin 2x-1$ at the point $x=0$ and $x=\pi$

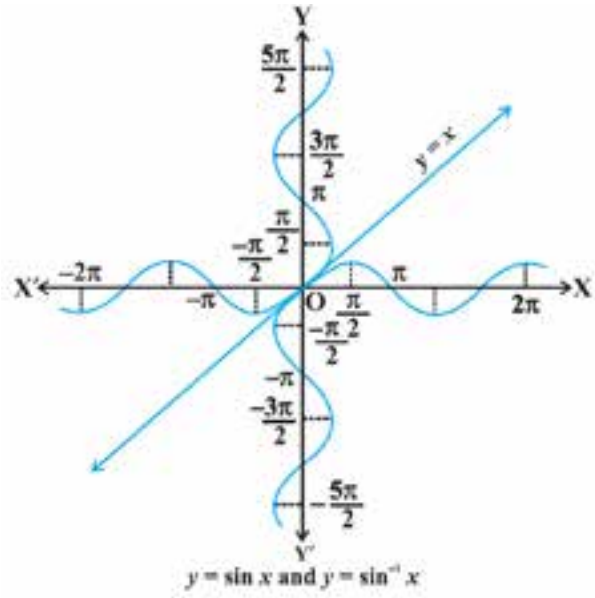
(a) Continuous at $x=0,\ \pi$

(b) Discontinuous at $x=0$ but continuous at $x=\pi$

(c) Continuous at $x=0$ but discontinuous at $x=\pi$

(d) Discontinuous at $x=0,\ \pi$

16. At which points does the graph of $y=\tan x$ intersect the X-axis?



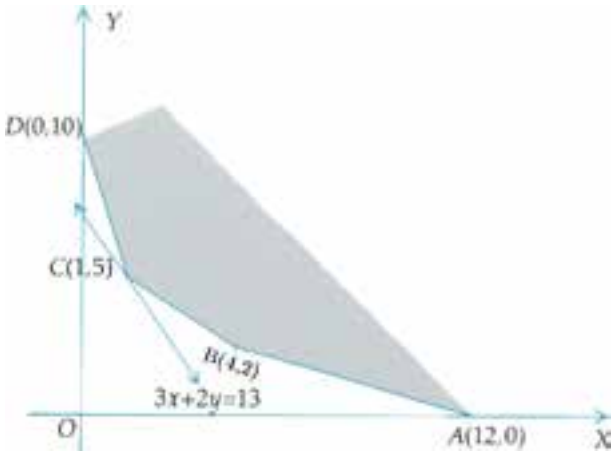
$y=\sin x$ and $y=\sin^{-1}x$

(a) $-\pi,-\pi/2,\pi/2,\pi$ (b) $-\pi,0,\pi$

(c) $-\pi/2,0\pi/2$ (d) $0,\pi/2\pi$

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17. The minimum value of $z = 3x + 2y$, if the feasible region for an LPP is as shown in Fig., is to be determined.



(a) 36 (b) 16

(c) 20 (d) 13

18. If $f(x) = \begin{cases} \frac{\log x}{x-1}, & \text{if } x \neq 1 \\ k, & \text{if } x = 1 \end{cases}$ is continuous at $x = 1$, then the value of k is

(a) 0 (b) -1

(c) 1 (d) e

19. **Assertion :** area of the parallelogram whose adjacent sides are $\hat{i} + \hat{j} - \hat{k}$ and $2\hat{i} - j + \hat{k}$ is $3\sqrt{2}$ square units.

Reason : area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $|\vec{a} - \vec{b}|$

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

20. **Assertion:** if $2P(A) = P(B) = \frac{5}{13}$ and $P\left(\frac{A}{B}\right) = \frac{2}{5}$, then $P(A \cup B)$ is $\frac{11}{26}$

Reason: J, F, E_1 and E_2 are two events. then

$$P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cup E_2)}{P(E_2)},\ 0 < P(E_2) \leq 1$$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Evaluate $\int \frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}} dx$.

OR

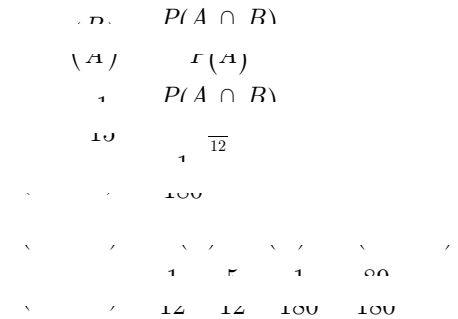
Evaluate $\int \frac{\cos\sqrt{x}}{\sqrt{x}} dx$.

22. Write the direction ratios of the vector $3\vec{a} + 2\vec{b}$, where $\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = 2\hat{i} - 4\hat{j} + 5\hat{k}$.

OR

Find the unit vector in the direction of the sum of vectors $2\hat{i} + 3\hat{j} - \hat{k}$ and $4\hat{i} - 3\hat{j} + 2\hat{k}$.

23. If $P(A) = \frac{1}{12}$, $P(B) = \frac{5}{12}$ and $P\left(\frac{B}{A}\right) = \frac{1}{15}$, then find $P(A \cup B)$



24. The x-coordinate of point on the line joining the points $P(2,2,1)$ and $Q(5,1,-2)$ is 4. Find its z -coordinate.

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25. Let R is the equivalence relation in the set $A = \{0,1,2,3,4,5\}$ given by $R = \{(a,b):2$ divides $(a-b)\}$. Write the equivalence class $[0]$.

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. If $\begin{vmatrix} 3x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 7 & 3 \end{vmatrix}$, then write the value of x.

27. Write the value of $\cos^{-1}\left(\frac{1}{2}\right) - 2\sin^{-1}\left(-\frac{1}{2}\right)$.

28. The volume of a sphere is increasing at the rate of 3 cubic centimetre per second. Find the rate of increase of its surface area, when the radius is 2 cm.

OR

Find the intervals in which the function

$$f(x) = \frac{3}{2}x^4 - 4x^3 - 45x^2 + 51 \text{ is}$$

(i) strictly increasing

(ii) strictly decreasing.

29. If $e^y(x+1) = 1$, then show that $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$.

30. If $\vec{a} = \hat{i} - \hat{j} + 7\hat{k}$ and $\vec{b} = 5\hat{i} - \hat{j} + \lambda\hat{k}$, then find the value of λ, so that $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular vectors.

31. Evaluate $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$.

OR

Evaluate $\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x} dx$.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Solve the differential equation

$$x \log|x| \frac{dy}{dx} + y = \frac{2}{x}\log|x|.$$

OR

Find the particular solution of the differential equation $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$ given that $y = 1$, when $x = 0$.

33. Maximize $Z = -x + 2y$, subject to the constraints:

$$x \geq 3,\ x + y \geq 5,\ x + 2y \geq 6,\ y \geq 0$$

OR

Maximize $Z = x + y$, subject to $x - y \leq -1$, $x + y \leq 0$, $x,\ y \geq 0$.

34. Evaluate $\int_0^{\pi} \frac{x}{a^2\cos^2 x + b^2\sin^2 x} dx$.

35. Find the values of p, so that the lines

$$l_1 : \frac{1-x}{3} = \frac{7y-14}{p} = \frac{z-3}{2} \text{ and}$$

$$l_2 : \frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$$

are perpendicular to each other. Also, find the equation of a line passing through a point $(3,2,-4)$ and parallel to line l_1.

SECTION - E

Case study based questions are compulsory.

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36. Rajneesh do outsourcing work for companies and runs a form processing agency. He collect form from different office and then extract data and record data on computer. In his office three employees Vikas, Sarita and Ishaan process incoming copies of a form. Vikas process 50% of the forms. Sarita processes 20% and Ishaan the remaining 30% of the forms. Vikas has an error rate of 0.06, Sarita has an error rate of 0.04 and Ishaan has an error rate of 0.03.



Based on the above information answer the following questions.

(i) The total probability of committing an error in processing the form.

(ii) The manager of the company wants to do a quality check. During inspection he selects a form at random from the days output of processed forms. If the form selected at random has an error, the probability that the form is not processed by Vikas.

37. A manufacturing company has two service departments, S_1, S_2 and four production departments P_1, P_2, P_3 and P_4.

Overhead is allocated to the production departments for inclusion in the stock valuation. The analysis of benefits received by each department during the last quarter and the overhead expense incurred by each department were:

Service Department	**Percentages to be allocated to departments**					
	S_1	S_2	P_1	P_2	P_3	P_4
S_1	0	20	30	25	15	10
S_2	30	0	10	35	20	5
Direct overhead expense ₹ '000	20	40	25	30	20	10



You are required to find out following using matrix method.

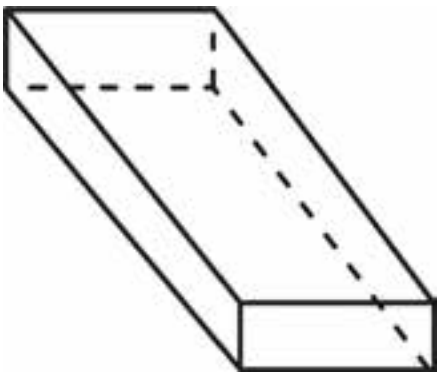
(i) Express the total overhead of the service departments in the form of simultaneous equations.

(ii) Express these equations in a matrix form and solve for total overhead of service departments using matrix inverse method.

(iii) Determine the total overhead to be allocated from each of S_1 and S_2 to the production department.

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38. Parallelepiped is the Greek word, which essentially means the object that has a parallel plane. Principally, the Parallelepiped is framed by the six parallelogram sides which bring about the prism or the 3D figure, and it consists of the parallelogram base. It can be categorized as anything but the polyhedron, where 3 sets of the parallel faces are made to combine for framing a three-dimensional (3D) shape that has six faces. The cube, cuboid, and rhomboid are the three exceptional cases. The Rectangular Parallelepiped consists of six faces in a rectangular shape.



The sum of the surface area of a rectangular parallelopied with sides of x, $2x$ and $\frac{x}{3}$ and a shape of radius of y is given to be constant.

On the basis of above information, answer the following questions.

(i) If S is the constant, then find the relation between S, x and y.

(ii) If the combined volume is denoted by V , then find the relation between V, x and y.

(iii) Find the relation between x and y when the volume V is minimum.

OR

(iv) If at $x = 3y$, volume V is minimum, then find the value of minimum volume and the value of S.

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